EPIDEMIOLOGY
OF INFECTIOUS DISEASES

# SCIENTIFIC SEMINAR ON INFECTIOUS DISEASES

Brussels, 22 May 2025

Sciensano
Epidemiology and public health
Epidemiology of infectious diseases
Rue Juliette Wytsmanstraat 14 | 1050 Brussels | Belgium

Scientific seminar on infectious diseases | 22 May 2025|
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## PROGRAM

| | |
|---|---|
| 08:30 | Registration with walking breakfast / Visit of stands |
| | **SESSION 1**<br>Cécile Meex (ULiège) & Heidi Theeten (Department Zorg) |
| 09:00 | Welcome address |
| 09:05 | Invasive bacterial infections: surveillance before and after covid – L. Cuypers (UZLeuven) |
| 09:30 | Vaccine hesitancy: presentation of Belgian data – G. Hendrickx (UAntwerp) |
| 09:55 | Mpox: evolution of the situation in 2024-2025 – L. Liesenborgh (ITM) |
| 10:25 | *Pertussis*: update on microbiological and epidemiological situation and maternal vaccination – H. Martini (UZ Brussels) & I. Peeters (Sciensano) |
| 10:50 | Coffee break / Visit of stands |
| | **SESSION 2**<br>Marcella Mori (Sciensano) & Elizaveta Padalko (UGent) |
| 11:20 | Bird Flu virus in humans and animals: update of the situation – A. Parys (Sciensano) |
| 11:45 | Air pollution in Europe and its impact on infectious diseases – O. Vandenberg (ULB) |
| 12:10 | Air sampling for surveillance of respiratory pathogens – C. Geenen (KU Leuven) |
| 12:30 | Lunch *and poster session* |
| | **SESSION 3**<br>Giulia Zorzi (UClouvain) & Wesley Mattheus (Sciensano) |
| 14:00 | *Major Salmonella* Enteriditis outbreak linked to egg consumption, Belgium, 2024-2025 – W. van Dyck (Sciensano) |
| 14:10 | Real place of highly multiplex PCR for gastro-intestinal pathogens – M. A. Argudin (UCLouvain) |
| 14:35 | Emerging threats of vector-borne diseases in Europe and worldwide, what is the risk for Belgium – E. Bottieau (ITM) |
| 15:05 | Respiratory syncytial virus: Preliminary analysis on the impact of nirsevimab – S. Blumental (ULB) & L. Demot (Sciensano) |
| 15:35 | Vaccine-derived polio virus 2 detection in wastewater in multiple European cities - K. Hansford (Sciensano) |
| 15:45 | Closing address and end of seminar |

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**1.** Mortensen JE, et al. Comparison of time-motion analysis of conventional stool culture and the BD MAX™ System Enteric Bacterial Panel (EBP). *BMC Clin Pathol.* 2015;15:9. **2.** Hirvonen JJ, et al. Comparison of BD Max™ System Cdiff and GenomEra C. difficile molecular assays for detection of toxigenic Clostridium difficile from stools in conventional sample containers and in FecalSwabs. *Eur J Clin Microbiol Infect Dis.* 2015;34(5):1005-1009. **3.** Felder RA et al. J Lab Autom. 2014;19(5):468–73.

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# ABSTRACTS OF PRESENTATIONS

## LIZE CUYPERS

UZ LEUVEN

### BIOGRAPHY

Lize Cuypers is a postdoctoral researcher who currently coordinates all activities related to the 8 National Reference Centres (NRCs) of human microbiology that are hosted at UZ/KU Leuven, of which respiratory pathogens, fungal infections and invasive pneumococcal disease are examples. She is working actively to broadly roll-out next-generation sequencing approaches within the clinical and research laboratory, within this context she has been responsible for the coordination of the Belgian SARS-CoV-2 genomic surveillance consortium during the COVID-19 pandemic. She obtained her PhD in 2016 at KU Leuven in the field of virology, focusing on the genetic diversity of the hepatitis C virus. During her postdoctoral years at KU Leuven she broadened her scope to other blood-borne viruses, mosquito-borne viruses, and respiratory viruses. Since 2020, she joined the University Hospitals of Leuven in her current position and is also affiliated to KU Leuven's Laboratory of Clinical Microbiology. While initially trained in virology, her interests have widened to viral, bacterial and fungal infections in the context of NRC activities.

### INVASIVE BACTERIAL INFECTIONS: SURVEILLANCE BEFORE AND AFTER COVID

Containment measures during the COVID-19 pandemic were effective in reducing transmission of SARS-CoV-2 and also other pathogens transmitted via respiratory droplets, such as *Streptococcus pneumoniae* (*Spn*), *Streptococcus pyogenes* (*Spy*), *Haemophilus influenzae* (*Hi*), and *Neisseria meningitidis* (*Nm*). These pathogens are leading causes of invasive bacterial infections (IBIs) associated with high morbidity and mortality. Country-wide surveillance of all four bacteria is performed by National Reference Centres (NRCs) in Belgium. To evaluate the evolution of the incidence of the different IBIs, a study was set up between the NRCs, including data from 2017 to 2024.

Case numbers were severely impacted for all four bacterial pathogens during the COVID-19 pandemic in agreement to other countries, respectively a reduction of 45.1%, 59.1%, 69.8% and 80.5% in the yearly number of infections with *Hi*, *Spn*, *Spy* and *Nm*, from pre-COVID (2017-2019) to epidemiological year 2020-2021. Following lifting of non-pharmaceutical interventions post-COVID, again an increase in IBIs was noted towards the end of 2022. This increase was preceded by a period with limited exposure to respiratory pathogens and hence an anticipated immunity decrease. Exceptional high numbers of group A streptococcal infections were observed for 2022-2023, reflecting an increase of nearly 90% compared to 2020-2021, being an increase of 64% compared to pre-COVID. Also incidences of other IBIs increased again, with for *Nm* infections an increase of 71.4% from 2020-2021 to 2022-2023, although reflecting a decrease of 31.9% compared to pre-COVID. Increases of 65.3% for *Hi* and 62.6% for *Spn* were observed from 2020-2021 to 2022-2023, respectively +36.8% and +8.7% compared to pre-COVID. Following the rapid increase of *Spy* infections in 2022-2023, case numbers decreased again by 33.9% in 2023-2024, however still largely exceeding case counts observed pre-COVID (+45.5%). Likewise for *Hi*, the number of infections decreased again in 2023-2024 (-26.0%), nevertheless still almost 15.0% higher compared to pre-COVID. While *Nm* was most impacted during the pandemic, post-COVID it is the only pathogen for which still a lower number of infections was recorded compared to pre-COVID (-33.6%). In contrast to the other IBIs, the increase of *Spn* infections pursued in 2023-2024 (+10.7%), which resulted in 17.5% more infections compared to pre-COVID years.

Long-term surveillance programs of IBIs clearly show the impact of preventive measures during and after the COVID-19 pandemic in Belgium, although the four bacterial pathogens were impacted differently. While similar patterns for the four invasive diseases were expected based on anticipated post COVID- 19 immunity decrease, other factors like emergence of (new) bacterial clones, impact on carriage rates, affected age groups, different viral-bacterial interaction and transmission mode may play a role, and warrant further investigation.

**GREET HENDRICKX**

UANTWERP

## BIOGRAPHY

Since 2007 Greet Hendrickx has been working as a Senior Project Coordinator at the Centre for the Evaluation of Vaccination, Vaccinopolis, University of Antwerp. In this role, she manages the executive secretariat for two European expert boards: the Viral Hepatitis Prevention Board and the Adult Immunization Board.
In 2019, she became the team lead for the CONFIVAX team, which integrates multi-disciplinary research and expertise on vaccination attitudes. Her work actively engages in European projects focused on vaccination attitudes, trust, willingness, communication, and education.
Additionally, she is a key member of the European Hub of the Vaccine Confidence Project.

## VACCINE HESITANCY: PRESENTATION OF BELGIAN DATA

In response to the European Council's 2018 recommendation (2018/C 466/01) to enhance cooperation against vaccine-preventable diseases, the European Commission committed to regularly reporting on vaccine confidence across the EU. This initiative, aligned with the State of Health in the EU process, aims to monitor public attitudes toward vaccination. The Vaccine Confidence Project (VCP), based at the London School of Hygiene & Tropical Medicine with a European hub at the University of Antwerp, has conducted vaccine confidence surveys across all EU member states — including Belgium — in 2018, 2020, and 2021.

In line with these efforts, and as part of the operational objectives of the Flemish vaccination health goals ("Gezondheidsdoelstellingen Vaccinatie") presented at the 2024 Health Conference of the Flemish Department of Health-Departement Zorg, a comparable regional study was conducted in Flanders. This cross-sectional survey assessed vaccine confidence among 1,500 participants: 1,000 adults from the general population (aged >18 years) and 500 parents of children aged ≤13 years.

Vaccine confidence was measured using the standardized Vaccine Confidence Index™ (VCI), enabling comparability with previous EU-wide studies. Statistical regression analyses were performed to identify associations between vaccine confidence and various sociodemographic and contextual predictors.

The findings provide a valuable baseline for monitoring vaccine confidence in Flanders and contribute to the broader European dataset. These results can inform targeted public health strategies and communication efforts. To deepen understanding of evolving trends and the underlying drivers of vaccine confidence, longitudinal studies — ideally incorporating qualitative methods — are recommended. Such research will be essential to support evidence-based policymaking and to foster sustained public trust in vaccination programs.

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## LAURENS LIESENBORGHS

ITM

### BIOGRAPHY

Laurens Liesenborghs is an infectious disease specialist and researcher specialized in emerging infectious diseases and humanitarian medicine. He has a diverse scientific background in key areas of infectious diseases, including preclinical drug development, virology, vaccinology, microbiology, clinical trials, and epidemiology.

During his PhD, he developed a novel preclinical model to study the pathophysiology of infective endocarditis. His focus later shifted to emerging viruses, beginning with his work for Doctors Without Borders on a Lassa Fever project in Sierra Leone. He then conducted postdoctoral research at the BSL3+ lab at KU Leuven, where he contributed to the development of a hamster model for COVID-19 and worked on testing antivirals and vaccines for SARS-CoV-2.

Currently, Laurens is an assistant professor at the Institute of Tropical Medicine and at KU Leuven. His research primarily focuses on mpox. He has been co-leading a major project on mpox transmission in the Democratic Republic of the Congo, where his team discovered the Clade Ib variant responsible for the 2024 multicounty outbreak. In Belgium, he is conducting several studies on mpox, including on asymptomatic presentations, high-risk contacts, and diagnostics, as well as participating in clinical trials on therapeutics. His broader research includes outbreaks and hemorrhagic fevers in Africa.

### MPOX: EVOLUTION OF THE SITUATION IN 2024-2025

In 2022, the mpox virus, previously confined to forested areas in Central and West Africa, gained notoriety after causing a worldwide outbreak. However, for those working in the field, the emergence of this virus was not entirely unexpected. Since the 1980s — following the eradication of smallpox and the discontinuation of smallpox vaccination campaigns — mpox has gradually been emerging, particularly in the Democratic Republic of the Congo (DRC). In this region, the virulent Clade I mpox virus circulates in an unknown animal reservoir and causes sporadic zoonotic outbreaks, which have been steadily increasing over the past four decades. Simultaneously, a subclade of the West African Clade II resurfaced in Nigeria in 2017. This strain, designated clade IIb, later spread internationally, culminating in the 2022 global Clade IIb outbreak. Just as this Clade IIb outbreak subsided, a novel variant of Clade I (subclade Ib) emerged in the DRC, sparking a second, sustained international outbreak marked by human-to-human transmission. In this presentation, Laurens Liesenborghs provides an overview of the past, present, and future of mpox, drawing on research conducted at ITM in collaboration with Congolese partners.

## HELENA MARTINI

UZ BRUSSELS

### BIOGRAPHY

Helena Martini is a molecular biologist at the Department of Microbiology of the Universitair Ziekenhuis Brussel, under Prof. D. De Geyter. Her main focus lies on her work for the National Reference Centres for *Bordetella pertussis* and for toxigenic corynebacteria. Within the NRC for *Bordetella pertussis*, she is responsible for the molecular typing of collected strains.

## *PERTUSSIS:* UPDATE ON MICROBIOLOGICAL AND EPIDEMIOLOGICAL SITUATION AND MATERNAL VACCINATION


Helena Martini[1], Oriane Soetens[1], Eveline Van Honacker[1], Isabelle Desombere[2].

1 Department of Microbiology, National Reference Centre for *Bordetella pertussis*, Universitair Ziekenhuis Brussel, Vrije Universiteit Brussel (VUB), Laarbeeklaan 101, 1090 Brussels, Belgium.
2 Laboratory of Immune Response, Scientific Direction Infectious Diseases in Humans, Sciensano, Brussels, Belgium.


After a three-year period of extremely low *pertussis* incidence due to COVID-19-related non-pharmaceutical interventions, 2023 and 2024 saw a major resurgence. The National Reference Centre (NRC) for *Bordetella pertussis* reported 1,048 cases in 2023 and 3404 in 2024, the latter being an almost five-fold increase compared to pre-COVID 2019.
Serodiagnosis accounts for less than half of these cases (48% in 2023, 37% in 2024), the rest was confirmed by PCR on respiratory samples. The NRC also attempts to culture these in order to collect strains for further characterization: 95 isolates were collected in 2023 and 293 in 2024.
Before the COVID-19 pandemic, circulating *B. pertussis* strains were increasingly lacking in pertactin expression, a possible effect of selective pressure against the acellular *pertussis* vaccine (aP). This phenomenon was seen not only in Belgium but world-wide. Virulence typing of the 2023 strains has revealed that all the new strains express pertactin again.
Another important observation is the appearance of macrolide resistant strains. Up until 2023, all collected strains were susceptible to erythromycin, but in 2024, we see three separate cases of erythromycin resistance. Macrolide resistant strains have also started appearing in other European countries, a worrying prospect as it may mirror the situation in China and other Asian countries, where macrolide resistant strains have become predominant over the past decade.
While the Belgian *pertussis* peak has calmed down considerably since the second half of 2024, it is important we remain vigilant and continue to monitor the epidemiological situation and strain characteristics.

## ILSE PEETERS

SCIENSANO

### BIOGRAPHY

Dr. Ilse Peeters is a general practitioner with a postgraduate degree in Tropical Medicine and International Health. She has five years of clinical experience as a physician in Belgium and has completed several infectious diseases internships abroad. Since 2019, she has been working at Sciensano in the Department of Epidemiology of Infectious Diseases. Her current focus is on vaccine-preventable diseases, specifically *pertussis* and the pediatric surveillance network, PediSurv.

## BELGIUM'S REGIONAL DISPARITIES IN MATERNAL VACCINATION COVERAGE: DO WE SEE REAL-LIFE IMPACT IN INFANT *PERTUSSIS* CASES?

I. Peeters[1], N. Hammami[2], C. Boulouffe[3], A. Taame[4], L. Cornelissen[1].

1 Sciensano, Belgium.
2 Departement Zorg, Belgium.
3 Agence Wallonne pour une vie de qualité, Belgium.

Background: Since mid-2023, Europe experiences intense *pertussis* circulation. Routine infant vaccination and vaccination of women during each pregnancy are key to protect the risk group of young infants. Most recent available data (2019) indicates 98% vaccination coverage for 3rd dose of *pertussis* vaccine in all Belgian regions. In contrast, maternal vaccination uptake differs by region, ranging from 85% in Flanders to 39% in Wallonia and 31% in Brussels. We hypothesize that this difference in maternal vaccination coverage could be reflected in infant case numbers.

Method: Our study period is from January until September 2024. Data on *pertussis* cases in <1y of age was extracted from the regional mandatory notifications (MN). Estimated incidence rates were calculated per region, using official population statistics (Statbel). Incidences in different regions were compared using incidence rate ratios (IRR) with 95% confidence intervals (CI).

Results: There were 422 confirmed *pertussis* notifications, of which 42% occurred in Flanders (n=177), 37% in Wallonia (n=157) and 20% in Brussels (n=86). This equals estimated incidence rates of 282/100,000 population in Flanders, 461/100,000 for Wallonia and 626/100,000 in Brussels. The IRR of Brussels *vs.* Flanders is 2.22 [95% CI 1.71-2.87], indicating a two-fold higher risk of *pertussis* infection for infants in Brussels compared to Flanders. The IRR of Wallonia *vs.* Flanders is 1.63 [1.32-2.03]. Since we use different organized sources for case numbers in the different regions, we cannot rule out a measurement bias. However, notification rates in adults are similar across the 3 regions and higher case ascertainment rates in Brussels and Wallonia are not known for other mandatorily notifiable pathogens.

Conclusion: Estimated incidence of *pertussis* in young infants is significantly higher in Brussels and Wallonia, compared to Flanders, which could be due to differences in maternal vaccination. Efforts to identify and address barriers to maternal vaccination should be made to improve coverage in all regions.

Keywords: *Pertussis* ; maternal vaccination ; infant *pertussis* cases ; vaccination coverage ; maternal vaccination coverage.
Presented at Belgian Pediatric Conference 2025, Mons.

## ANNA PARYS

SCIENSANO

### BIOGRAPHY

Anna Parys obtained her master in Veterinary Medicine (main subject research) at Ghent University in 2018. Thereafter, she started her PhD with a focus on pig as a model for development of broadly protective influenza A vaccines and public health implications of swine influenza. End of 2022, Anna obtained her PhD at the Laboratory of Virology, Faculty of Veterinary Medicine, Ghent University. At the same time, Anna started working at Sciensano, first at the Experimental Center in Machelen where she gained experience on the coordination and supervision of large animal (GLP) experiments for clients. Thereafter, since January 2024, Anna started working as a scientist at the NRC Respiratory pathogens. Thanks to her background on zoonotic influenza, she is currently leading a project on animal influenza transmission to humans, called ZOOIS and participated as an expert in the ECDC External Expert Panel on Avian Influenza.

### BIRD FLU VIRUS IN HUMANS AND ANIMALS: UPDATE OF THE SITUATION

Influenza is an enveloped RNA virus that can cause acute respiratory disease in mammals and humans. In most cases, influenza viruses are species-specific, yet occasionally transmission from one species to another or to humans can occur. Subsequently, if this virus has the ability to spread efficiently in the human population, an influenza pandemic might occur as was the case in 1918, 1957, 1968 and 2009.

In birds, the highly pathogenic H5N1 strain emerged in commercial geese in Asia around 1996, and spread in poultry throughout Europe and Africa in the early 2000s. Ever since, this avian influenza virus has dominated outbreaks around the world, including the current ones. Most recently, in April 2024, an outbreak of the highly pathogenic avian influenza (HPAIV) H5N1 was reported in dairy cattle in Texas, with high virus concentrations detected in raw milk. This outbreak led to sporadic transmission events to humans and various mammalian species and resulted in spillback events into both domestic and wild avian species. In Belgium, surveillance of seasonal influenza viruses in humans is organized through several sentinel networks. Besides, for avian influenza, active (poultry) and passive (wild birds) surveillance are in place. Yet, surveillance of influenza viruses at the animal-human interface was not yet implemented, which is why the clinical pilot study ZOOIS was invented. During this pilot study, several dedicated sentinel networks among at-risk workers were initiated (ZOOIS-sentinel) together with the second part of the study (ZOOIS-outbreak) which aims at implementing virological and serological follow-up among people exposed to potential zoonotic influenza. Among them the recent outbreaks of H5 in Belgium were investigated.

## OLIVIER VANDENBERG

ULB

### BIOGRAPHY

Olivier Vandenberg has been trained as Clinical Microbiologist (2001) and completed his PhD in Biomedical Sciences (2006) at Université Libre de Bruxelles (ULB), Brussels, Belgium.
From 1996 to 2024, he held various positions at the University Laboratory of Brussels (LHUB-ULB).
Since 2023, he took the lead of the Environmental and Occupational Health Research Centre of the School of Public Health, ULB, Brussels, Belgium. In addition, he is currently Professor of Microbiology in the School of Public Health and in the Faculty of Medicine of the Université Libre de Bruxelles. Most of his research focuses on emerging pathogens and the clinical impact of new diagnostic tools in industrialised countries and low-resource countries, before turning to the study of the impact of environmental changes on the incidence of infectious diseases. Since 2000, he has also supervised the consolidation of several microbiology laboratories allowing the implementation of infectious disease surveillance program in industrialized but also in Low and Middle Incomes Countries (LMICs). Besides this, he collaborates with different manufacturers in the development of new approaches for the diagnosis and control of infectious diseases.

## AIR POLLUTION IN EUROPE AND ITS IMPACT ON INFECTIOUS DISEASES

Air pollution, particularly from industrial emissions, vehicle exhaust, biomass burning, and chemical pollutants, poses a substantial threat to public health. Recent research has established a growing link between exposure to polluted air and an increased risk of infectious diseases, especially respiratory infections. This relationship is multifaceted and involves both direct and indirect biological, environmental, and societal factors.

At the core of this link is the impact of air pollution on the human respiratory system. Pollutants such as particulate matter (PM2.5 and PM10), nitrogen dioxide ($NO_2$), sulfur dioxide ($SO_2$), ozone ($O_3$), and carbon monoxide (CO) injure the epithelial lining of the airways making individuals more susceptible to infections like influenza or pneumonia.



Air pollution also plays a role in the transmission dynamics of infectious diseases. Studies have shown that viruses such as SARS-CoV-2 can adhere to air particles, potentially increasing their persistence in the atmosphere and facilitating wider spread, especially in densely populated urban areas.

Furthermore, pollution-driven climate change indirectly enhances the spread of infectious diseases by altering ecosystems. For instance, warmer temperatures and shifting weather patterns can expand the habitat ranges of disease-carrying vectors, increasing the risk of vector-borne diseases.

Air pollution also contributes to urban heat islands and changes in humidity that may affect the viability and transmission of airborne pathogens.

This lecture will summarize the main challenges posed by air pollution in the prevention and control of infectious diseases in Europe but will also outline the various research perspectives that these changes are giving rise to.

## CASPAR GEENEN

KU LEUVEN

### BIOGRAPHY

Caspar Geenen is a PhD researcher at KU Leuven with a focus on public health and infectious diseases. He graduated :*cum laude* with a Master of Medicine (MD) from KU Leuven in 2013 and initially pursued a career in ophthalmology in the United Kingdom, working as a Specialty Doctor at Sunderland Eye Infirmary. During the COVID-19 pandemic, after volunteering to help manage an outbreak in a nursing home, he shifted his focus to public health and infectious diseases. In 2021, he joined KU Leuven's Laboratory of Clinical Microbiology as a research assistant and coordinator of the university's COVID-19 contact tracing team. Since 2022, he has been working on his PhD, supervised by Prof. Dr. Emmanuel André. He investigates new approaches for outbreak investigation and control, with a particular interest in indoor air surveillance and digital tools for contact tracing.

### AIR SAMPLING FOR SURVEILLANCE OF RESPIRATORY PATHOGENS

Tracking the number of infections in a population is crucial for optimising outbreak control measures. Conventionally, epidemiological indicators rely on healthcare data, such as positive tests or clinical diagnoses like influenza-like illness (ILI) or severe acute respiratory infection (SARI). However, COVID-19 exposed key limitations, such as testing biases, delays, and underrepresentation of subclinical infections.

Environmental sampling has since gained renewed interest to obtain more accurate measures of disease circulation. While wastewater sampling is now well-established, air sampling in community settings has shown promise, offering some unique potential benefits (1). For example, air samples are relatively uncontaminated, making them suitable for metagenomic analysis (2). They can target specific age groups or building types, providing insights on demographic groups and settings where diseases are spread or prevalent. Crucially, air samples may provide more timely epidemiological indicators, especially for pathogens transmitted via the airborne route. These pathogens are shed in the air as soon as a case becomes infectious, and their concentration in the air is dependent on outbreak control measures, such as face masks. Therefore, air samples may represent not just the number of cases, but the level of exposure of the population, resulting in responsive epidemiological signals.

Our research focuses on molecular testing of air samples collected with high-airflow impaction samplers. We have shown that genetic material of respiratory pathogens could be found in most samples from a range of community settings (1). We also showed the important effect of ventilation in reducing airborne pathogen concentrations.

In a childcare setting, we compared air samples to individual respiratory tract samples, and found that air sampling could rapidly detect outbreaks and reveal distinct pathogen shedding patterns (3).

The broader the “catchment area” of an air sample, the greater its epidemiological value. Scaling up can be achieved by sampling from central ventilation systems or by collecting samples over longer periods (4). Both approaches have already shown promise in real-world studies .

In summary, air sampling is emerging as a valuable tool for respiratory pathogens surveillance. With further validation, it could offer timely, sensitive indicators to complement other epidemiological signals.

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## WOUTER VAN DYCK

### SCIENSANO

### BIOGRAPHY

Wouter van Dyck is a scientist in the team for the surveillance of food and water borne diseases of the Epidemiology of Infectious Diseases service at Sciensano since May 2024. He graduated from the masters "Infecious and Tropical Diseases" (2014) and "Epidemiology" (2023) at the University of Antwerp. In between these two masters, Wouter worked as a case report form designer in a clinical research organisation.

## MAJOR *SALMONELLA* ENTERITIDIS OUTBREAK LINKED TO EGG CONSUMPTION, BELGIUM, 2024-2025.

W. van Dyck[1], D. Van Cauteren[1], A. Christiaens[2], V. Cantaert[2], B. Verhaegen[3], L. Delbrassinne[3], G. De Muylder[1], A. Devos[4], N. Hammami[4], C. Broucke[4], W. Mattheus[5], V. Laisnez[1].

1 Department of Epidemiology and Public Health, Sciensano, Brussels, Belgium.
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### Background

On 20 January 2025, the Belgian National Reference Center (NRC) reported a genomic cluster of *Salmonella* Enteritidis cases. The increase started in late 2024 with 21 cases reported in early 2025. Cases were predominantly located in two geographical clusters in Flanders and showed an unusual age distribution (more adults than expected). Therefore Sciensano initiated an outbreak investigation together with the regional health authorities.

### Methods

The investigation team was composed of experts from Sciensano, Departement Zorg and the Federal Agency for the Safety of the Food Chain (FASFC). Outbreak cases were defined as *S* Enteritidis infections within the identified genomic cluster and isolation date after 11 November 2024. To generate hypotheses, we conducted exploratory interviews using an open-ended approach with the most recent cases (isolation date after 6 January 2025). Interviews focused on food consumption within three days before symptom onset, specific places of purchase, events, and whether someone around the case got symptoms.

## Results

By 4 March 2025, the NRC reported 98 outbreak cases of which 63 were eligible for questioning. We interviewed 51 cases and identified four independent places of exposure for which the traceability indicated eggs originating from the same farm. Consequently the FASFC imposed the diversion of the eggs to the egg processing industry. Samples from the laying hen farm (dust, faeces and eggs) tested positive for *S.* Enteritidis and the isolates matched the outbreak strain. Hence a recall was launched.

## Conclusion

Epidemiological and microbiological investigations confirmed the link between *S.* Enteritidis cases and eggs from a Belgian egg producer. The magnitude of this outbreak is likely to be underestimated given the number of eggs produced, their long shelf life and the fact that only the most severe cases were diagnosed and reported. A strong collaboration and information sharing between all stakeholders resulted in comprehensive measures taken to stop the outbreak.

## Keywords

*Salmonella* Salmonella Enteritidis ; foodborne outbreak ; whole genome sequencing ; trawling questionnaire ; eggs.

## MARIA ANGELES ARGUDIN

UC LOUVAIN

BIOGRAPHY

Maria A. Argudín, MSc, PhD, has been the scientific manager of the molecular microbiology lab at Cliniques Universitaires Saint-Luc (CUSL, Brussels) since June 2020. She received an extraordinary award and earned her master's degree in Biology from the University of Oviedo (UNIOVI, Spain) in 2004, followed by a Diploma of Advanced Studies in 2005. She lectured at UNIOVI until obtaining her European PhD (*Summa Cum Laude*) in 2011. Her research on the molecular epidemiology and genetic basis of resistance and virulence of *Staphylococcus aureus* was conducted at UNIOVI and the Federal Institute for Risk Assessment in Berlin. From 2012 to 2014, she completed a postdoc on staphylococci at CODA-CERVA (Brussels). She then served as chief expert scientist at the National Reference Centre for Staphylococci (LHUB-ULB, Brussels) from 2014 to 2020. With more than 20 years of experience in molecular epidemiology, including diagnostics, typing, microarray, sequencing, and bioinformatics, she has co-authored over 65 peer-reviewed papers. At CUSL, she is currently responsible for testing and validating molecular pathogen detection techniques and is involved in the National Reference Centres for *Bartonella* and viral hepatitis.

## REAL PLACE OF HIGHLY MULTIPLEX PCR FOR GASTRO-INTESTINAL PATHOGENS

Diarrhea is a common medical concern, with most cases caused by infectious agents — viruses being the most frequent, followed by bacteria and parasites. Laboratory diagnosis depends on clinical features: while many cases resolve spontaneously, testing is essential in severe, persistent, or high-risk patients. Traditional methods like culture, microscopy, and antigen tests are time-consuming, require trained personnel, and often yield low positivity rates. Highly multiplex PCR panels offer a promising alternative, enabling rapid, simultaneous detection of multiple pathogens with high sensitivity and specificity [1,2]. These panels significantly reduce turnaround time, improve workflow efficiency, and are cost-effective over time by decreasing unnecessary treatments, additional testing, and hospital stays [1,2]. However, they have limitations: target bias (only panel-included pathogens are detected), potential false positives or negatives, and clinical interpretation challenges [1,2]. Nucleic acid detection doesn't always indicate active infection; results may reflect past exposure or asymptomatic carriage [1,2]. *Ct* values can aid interpretation by indicating pathogen load [3,4]. For bacterial targets, reflex cultures remain essential for testing antimicrobial susceptibility [1,2]. While not perfect, multiplex PCR panels are a valuable tool in diagnosing gastrointestinal infections, enhancing pathogen detection and supporting faster, more informed clinical decisions — when interpreted with proper clinical and epidemiological context.

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## EMMANUEL BOTTIEAU

ITM

### BIOGRAPHY

Emmanuel Bottieau is a Belgian physician born in 1963, who gained clinical experience in various tropical fields. He specialized in Internal Medicine with a particular interest in Infectious and Tropical Diseases in 1999. He defended his PhD research ("Fever after a stay in the Tropics") at the University of Antwerp in 2007. He currently works as clinician, teacher, researcher and expert in the Travel and HIV/STD Clinics and in the Department of Clinical Sciences of the Institute of Tropical Medicine, Antwerp, Belgium. He was appointed in 2012 as Professor of Tropical Medicine, and is since then file-holder of the Tropical Medicine and Clinical Decision-Making course. His main research topics are (1) the integrated management of challenging syndromes (febrile illness, neurological disorders, skin infection) in the tropics and in travelers, (2) the diagnosis and treatment of (Neglected) Tropical Diseases (malaria, schistosomiasis, strongyloidiasis, leishmaniasis, trypanosomiasis, cysticercosis, monkeypox), and (3) the surveillance of travel-related pathology within European and global networks of travel clinics (TropNet, GeoSentinel).

### EMERGING THREATS OF VECTOR-BORNE DISEASES IN EUROPE AND WORLDWIDE, WHAT IS THE RISK FOR BELGIUM?

Climate change is deeply modifying the ecology of most pathogens worldwide, and this phenomenon is accelerated and amplified by the massive increase in travel and migration. Any new infectious disease emerging somewhere on the planet may be rapidly transported anywhere else, as exemplified by the successive epidemics which have hit large geographical areas in the past two decades.

In this talk, we will focus on the subgroup of vector-borne diseases, the most affected by those recent developments, that represents one of the major threats to global health. The vectors that may transmit pathogens mainly consist of mosquitoes, ticks, flies and midges. Several scenarios may contribute the introduction to, or expansion of "exotic" pathogens in, Belgium : (1) the progressive invasion of new areas by vectors and pathogens already endemic in Europe, with geographical expansion of disease endemicity; (2) the introduction of tropical pathogens to Europe by infected travelers, with the risk of secondary autochthonous transmission wherever competent vectors are present; and (3) the sporadic spillage of travel-related infections to Belgium from ongoing outbreaks or new endemic threats in tropical areas.



Several recent examples of increasing threats to Europe/Belgium will be presented during the talk. For scenario 1, the geographical spread of mosquito-borne West Nile virus (WNV) and tick-borne encephalitis virus (TBEV) and Crimean-Congo hemorrhagic fever virus (CCHFV) infections will be discussed, as well as the progression of tick-borne rickettsiosis or sandfly-borne leishmaniasis.  For scenario 2, the increasing number of autochthonous outbreaks of arboviral infections (such as dengue, zika or chikungunya viruses) in Southern Europe will be highlighted, in relation to the unstoppable expansion of invasive *Aedes albopictus*. For scenario 3, the recent emergence of travel-related midge-borne Oropouche virus (OROV) infection and artemisinin-resistant *Plasmodium falciparum* malaria will illustrate the extreme variety of individual threats and clinical challenges for European travelers and physicians. The optimal surveillance and control activities to consider for each respective scenario will be briefly discussed, as exhaustiveness is out of reach during a short presentation.

**SOPHIE BLUMENTAL & LAURANE DE MOT**

ULB - SCIENSANO

## BIOGRAPHY

Sophie Blumental is a pediatric infectious disease specialist and medical sciences PhD, graduated from ULB. Her PhD involved collaborations with several hospitals in Belgium and abroad (London, Paris, Bordeaux). She has done the majority of her career as ID consultant in Brussels academic public hospitals (especially Huderf) and works now in the Chirec network. She is also scientific collaborator at ULB, appointed expert for the Superior Health Council (vaccination section, co-chairs of pediatric RSV and pneumococcal prevention working groups) as well as active member of the RSV national task force.

Laurane De Mot is a bioengineer with a PhD in mathematical modelling of gene networks. She worked for 7 years as a bioinformatician in the pharmaceutical industry, focusing on vaccine and drug mode of action, before joining Sciensano during the COVID-19 pandemic. She contributed to COVID-19 hospital surveillance, then joined the Epidemiology of Respiratory Infections team, which she now leads. The team monitors SARS-CoV-2, influenza, RSV, and publishes a weekly bulletin on acute respiratory infections.

Laurane and Sophie collaborate to improve pediatric RSV surveillance in Belgium and assess the impact of new preventive measures.

## RESPIRATORY SYNCYTIAL VIRUS: PRELIMINARY ANALYSIS ON THE IMPACT OF NIRSEVIMAB

Since 2024, new tools for preventing pediatric RSV disease have become available in Belgium: a monoclonal antibody, Nirsevimab (Beyfortus®, Sanofi), reimbursed since October 2024, and a vaccine for pregnant women (Abrysvo, Pfizer), available from January 2024 but only reimbursed from January 2025. Accordingly, the vast majority of infants eligible for prevention in our country were immunized with Beyfortus® during this first season of RSV prevention.

In 2023, the SARI surveillance system (Severe Acute Respiratory Infections, based on 10 sentinel hospitals) was adapted to more comprehensively include pediatric RSV cases. In parallel, the RSVPed study was launched to broaden data collection on pediatric RSV-related hospitalizations, including 20 additional hospitals not participating in the SARI surveillance. Data from both studies covering the 2023–2024 (pre-prevention) and 2024–2025 (first year of prevention) seasons were analyzed in order to assess the impact of preventive measures on a national scale.

Preliminary results showed a substantial reduction of at least 35% in pediatric RSV-related hospitalizations during the 2024–2025 season compared to the previous one. A significant decrease in ICU admissions was also observed. Epidemiological changes mainly concerned infants under 6 months of age — the age group targeted by prevention and eligible for Beyfortus® reimbursement due to their higher risk of complications. Moreover, based on a test-negative design analysis using preliminary SARI data, the effectiveness of Nirsevimab was estimated at over 80%, with no waning of protection over time (similar effectiveness whether the children were immunized within the 60–90 days or the 2–30 days before infection). Finally, immunized infants who nonetheless required hospitalization due to RSV infection showed a less severe pattern of infection compared to non-immunized infants in the same age group.

In conclusion, the introduction of Nirsevimab has already had a marked impact on the pediatric RSV burden in Belgium. Further analyses on benefits for specific subgroups (*i.e*, children with comorbidities, born prematurely, or coinfected with other viruses) should still be carried out. Moreover, data on prevention uptake and acceptability among parents and healthcare workers are under investigation to help guide upcoming preventive campaigns.

**KIMBERLEY HANSFORD**

ULB

## BIOGRAPHY

Dr. Kimberley Hansford is a Scientific Collaborator at Sciensano, where she coordinates Belgium's national poliovirus surveillance and collaborates on the EU-WISH program. She has a background in infectious diseases epidemiology, laboratory science, and vaccine-preventable diseases control, with experience spanning vaccine development and public health policy. Her current work focuses on maintaining Belgium's polio-free status through integrated clinical and environmental surveillance strategies.

## VACCINE-DERIVED POLIO VIRUS 2 DETECTION IN WASTEWATER IN MULTIPLE EUROPEAN CITIES

Despite high vaccination coverage in many European countries, recent detections of vaccine-derived poliovirus type 2 (VDPV2) in urban wastewater highlight the continued relevance of environmental surveillance. This presentation will summarise findings from multiple cities and explore their implications for population immunity, early warning systems, and global eradication efforts. The results reinforce the need for sustained vigilance, even in regions where polio is considered eliminated.

# ABSTRACTS OF POSTERS

# A PHASE 3, RANDOMIZED TRIAL INVESTIGATING THE SAFETY, TOLERABILITY, AND IMMUNOGENICITY OF V116, AN INVESTIGATIONAL ADULT-SPECIFIC PNEUMOCOCCAL CONJUGATE VACCINE, IN PNEUMOCOCCAL VACCINE-NAÏVE ADULTS 18–64 YEARS OF AGE WITH INCREASED RISK FOR PNEUMOCOCCAL DISEASE.

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10 Sky Integrative Medical Center/SKYCRNG, Ridgeland, Mississippi, USA.
11 MSD Belgium, Brussels, Belgium) on behalf of the authors.

## Background

Adults with underlying chronic medical conditions are at increased risk of pneumococcal disease (PD). V116 is an investigational, 21-valent, adult-specific pneumococcal conjugate vaccine (PCV) containing the most prevalent serotypes (STs) associated with PD in adults from regions with established paediatric vaccination programs. The Phase 3 STRIDE-8 study (NCT05696080) evaluated the immunogenicity, safety and tolerability of V116 in adults 18–64 years of age at increased risk of PD.

## Methods

Pneumococcal vaccine-naïve participants with ≥1 underlying chronic medical conditions (*i.e.* diabetes *mellitus*, heart, kidney, liver, and lung disease) were randomized 3:1 to receive one dose of V116 or 15-valent PCV (PCV15) on Day 1 followed by placebo or one dose of pneumococcal polysaccharide vaccine (PPSV23), respectively, at Week 8. Safety was evaluated as the proportion of participants with adverse events (AEs). Immunogenicity was assessed by serotype-specific opsonophagocytic activity (OPA) geometric mean titers (GMTs) and immunoglobulin G (IgG) geometric mean concentrations (GMCs) at baseline (Day 1) and 30 days post-vaccination (Day 30 for V116+placebo and Week 12 for PCV15+PPSV23).

## Results

Of 518 participants randomized, 516 were vaccinated and received either V116 (n=386) or PCV15 (n=130) on Day 1; 96.7% completed the trial. One or more AEs occurred in 265 (68.7%) and 118 (90.8%) participants vaccinated with V116+placebo or PCV15+PPSV23, respectively. V116 was immunogenic for all 21 STs based on OPA GMTs, with comparable responses to PCV15+PPSV23 for the 13 STs common to V116 and PCV15+PPSV23, and higher responses for the eight STs unique to V116. IgG GMCs were consistent with OPA GMTs.

## Conclusion

V116 is well tolerated and immunogenic in adults 18–64 years of age at increased risk of PD, with comparable immune responses to PCV15+PPSV23 for common STs and higher immune responses for unique STs. These findings support V116 as a novel population-specific vaccine for the prevention of PD in adults with chronic medical conditions at increased risk of PD.

## Keywords

Vaccines ; clinical trial ; pneumococcal disease ; adult ; chronic medical conditions.

Presented at IDweek 2024, Los Angeles, USA.

# DATA LINKAGES FOR IMPROVED INFECTIOUS DISEASE SURVEILLANCE IN BELGIUM: A PROTOCOL FOR THE EPI-LINK PROJECT.

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## Background

The COVID-19 pandemic highlighted the importance of data linkage on an individual level to set up an efficient epidemiological surveillance of COVID-19. As such, direct linkages were established between laboratory test results, vaccination status, disease outcome (hospitalization, death) and clinical and administrative records. This approach can be extended to other infectious diseases, including newly emerging pathogens, to improve both continuous surveillance and pandemic preparedness with the aim to steer protective measures to improve population health. The implementation of the EPI-LINK project for routine surveillance or in the event of a Public Health Emergency (PHE) is described here.

## Methods

The EPI-LINK project integrates existing data collections through individual linkage using a pseudonymized identifier based on the unique national registry number. It involves linking infectious disease data (*e.g* laboratory test results and clinical hospital data) with other health and administrative data, including vaccination registers, healthcare consumption data, mortality data, and socio-economic and demographic information. This elaborate linkage comes with a number of legal, technical and logistical challenges to address.

## Results

EPI-LINK allows to build an infrastructure that will enable more comprehensive analyses to be carried out for improved surveillance of infectious diseases in Belgium, either in routine circumstances or in the event of a PHE. The expected outcomes include, but are not limited to: having more complete accurate data on vaccination status for cases, monitoring cause-specific mortality for specific pathogens using a proxy, more accurately estimating the number of cases via capture/recapture methods, assessing vaccine uptake, safety, and effectiveness in real-world settings.

## Conclusion

The EPI-LINK project will reinforce the continuous surveillance of infectious diseases through linkage of different databases at an individual level. The linkage of datasets can be expanded to allow for the collection and analysis of more granular data, which can support public health decision-making during a PHE.



## Keywords

data linkage ; health registry ; infectious diseases ; surveillance ; pandemic preparedness.

# KNOWLEDGE AND PERCEPTION OF TICK EXPOSURE AND LYME BORRELIOSIS IN THE GENERAL POPULATION IN BELGIUM.

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4 Pfizer, Paris, France.

## Background
Lyme borreliosis (LB), a bacterial infection caused by various genospecies of *B. burgdorferi s.l.* complex, is transmitted to humans through the bite of infected *Ixodes* spp ticks and is the most common tick-borne disease in Europe. In Belgium, data about the general population's knowledge and risk perception towards LB are limited. In 2022, we surveyed adults in 20 European countries to investigate awareness about ticks and LB and level of concern about contracting LB. This abstract presents the results of the survey for respondents in Belgium.

## Methods
We used an existing survey panel to conduct an online survey of adults aged 18-65 years old, with recruitment quotas on age, gender, and region. The survey included questions about LB knowledge, perception of tick exposure, and outdoor activities. We conducted descriptive analyses with weighting to adjust for the complex survey design.

## Results
Of 1224 respondents, 93% were aware of ticks and 84% were aware of LB. Among respondents that were aware of ticks and LB, 75% considered LB a severe disease, 39% were concerned about contracting LB, and 25% perceived themselves at risk of contracting LB. Overall, 32% of respondents reported a tick bite at least once in their lifetime. Always or often conducting tick checks was reported by 32% of respondents. Respondents reported spending an average of 8 hours outdoors at home and an average of 9 hours outdoors away from home each week.

## Conclusion
In Belgium, there is a broad knowledge of ticks and LB in the general population. Because people spend considerable time outdoors and do not regularly use prevention measures, new interventions are needed to address the risk for contracting LB and to mitigate this important public health concern.



## Keywords
Lyme borreliosis ; Ticks ; survey ; awareness ; risk perception.

# THE POWER OF LONGITUDINAL WASTEWATER METAGENOMICS TO TRACK GENOMIC AND CLINICAL EPIDEMIOLOGY OF HUMAN-ASSOCIATED VIRUSES.

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2 Department of Laboratory Medicine, University Hospitals of Leuven, Leuven, Belgium.

## Background

Environmental surveillance has re-emerged as a crucial tool for monitoring viral epidemiology. Traditional wastewater-based approaches rely on targeted assays such as qPCR panels, requiring prior knowledge of circulating viruses. To overcome these limitations, we employed a comprehensive metagenomics-based hybrid-capture approach targeting over 15,000 viral strains combined with short-read next-generation sequencing (NGS).

## Methods

We collected weekly wastewater samples from a treatment plant in Leuven, Belgium, over 122 weeks (~2.5 years). Using hybrid-capture with NGS and analyzing our results with a reference-guided assembly pipeline, EsViritu, we identified viral genomes against a database of all human and animal viruses in GenBank. On average, we recovered 110 viral genomes (or segments) per sample, using a 50% horizontal coverage cut-off. Most reads mapped to genomes belonging to the *Polyomaviridae, Parvoviridae, Adenoviridae, Circoviridae,* and *Sedoreoviridae*.

## Results

Using our comprehensive genomic data, we focused on clinical and genomic surveillance of important human pathogens. By comparing wastewater-derived rotavirus reads with Belgium's nationwide clinical cases, we observed a strong correlation supporting wastewater surveillance for monitoring viral trends. Similarly, SARS-CoV-2 reads from wastewater closely followed local case numbers at a university hospital, confirming its applicability for clinical epidemiology. Moreover, norovirus genotyping revealed a shift from GII.4 in 2022–2023 to GII.17 in 2024, mirroring trends in the recent literature reported in the US, Austria, France, Germany, and Ireland.



## Conclusion

Overall, our study demonstrates that longitudinal wastewater metagenomics enables real-time genomic and epidemiological surveillance of multiple viruses at once, proving itself as a strong public health monitoring tool.

## Keywords

environmental surveillance ; wastewater surveillance ; viral metagenomics ; metagenomics ; genomic surveillance.

# UNTAPPED POTENTIAL OF WASTEWATER FOR ANIMAL AND POTENTIALLY ZOONOTIC VIRUS SURVEILLANCE: PILOT STUDY TO DETECT NON-HUMAN ANIMAL VIRUSES IN URBAN SETTINGS.

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3 Department of Laboratory Medicine, University Hospitals of Leuven, Leuven, Belgium.

## Background

Wastewater surveillance has become an essential tool for monitoring viral outbreaks and surveillance of human viruses. While PCR-based methods are most frequently used, more advanced techniques, such as shotgun metagenomics in combination with viral capture methods, have been developed. In this study, we focus on tracking animal specific and zoonotic viruses in city wastewater using metagenomics.

## Methods

We collected 6 wastewater samples from Leuven and Brussels, situated in the center of Belgium. Automated wastewater samplers collected 50 mL samples every 10 minutes resulting in a 24h composite influent wastewater. All samples were processed using the TWIST comprehensive research panel capture method, designed to target over 3,000 human and animal viruses and 15,000 strains. Sequencing was performed on the AVITI sequencing platform, targeting an average of ten million reads per sample. The sequencing data were analyzed using the EsViritu tool.

## Results

Over 2,294 viral genomes or segments were recovered from six wastewater samples. Of these, 168 were associated with non-human vertebrate animals, including cats, dogs, pigeons, and rats, spanning 51 virus species. We identified near-complete genomes of clinically relevant animal viruses, such as pigeon circovirus, chicken anemia virus, feline bocaparvovirus 2, canine minute virus, rat coronavirus, canine parvovirus, and porcine circovirus. Additionally, we noted the presence of viruses with known cross-species transmission potential, including porcine torovirus, rosavirus, hepatitis E virus, rat hepatitis virus, and cardiovirus.

## Conclusion

The results demonstrate the ability to track a wide range of animal viruses in urban wastewater, potentially forming an early warning system for zoonotic diseases, ultimately being a useful tool for One Health based public health approaches.

## Keywords

animal virus surveillance ; zoonosis ; hepatitis E ; wastewater surveillance ; environmental surveillance.

## HOTSPOT OR NOT? A STATISTICAL COMPARISON OF DENGUE SURVEILLANCE METHODS.

N. Yahaghi[1], A. Juache[2], G. Hendrickx[2], C. Marsboom[2], S-N. Sharma[2], R. Srivastava[3], S Dhawan[3], B. J. Raju[4], R. Sharma[4], R. Theunissen[5], J. Broeckx[5], H. Rehman[5], P. Prabhakaran[4].

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### Background & Aims

This study evaluates four hotspot detection methods for dengue fever in Lucknow, Uttar Pradesh, India, using georeferenced cases from 2020 to 2022.

### Methods

The retrospective cross-sectional analysis included 845, 1,762, and 3,610 confirmed cases from public and private hospitals during weeks 35 to 52. The study compared Kernel Density Estimation (KDE), kriging, local Moran's I (LMI), and spatial scan statistics against Getis-Ord (GO) Gi* as a reference, using classic and spatial fuzzy Kappa statistics.

### Results

Results revealed distinct spatial patterns: KDE and scan statistics identified differing hotspots, while kriging and LMI closely matched GO Gi*. LMI had the highest agreement (0.66 ± 0.13), specificity (0.99 ± 0.01), and accuracy (0.69 ± 0.12). Kriging followed with agreement (0.48 ± 0.05), specificity (0.97 ± 0.03), accuracy (0.56 ± 0.07), and slightly higher sensitivity (0.43 ± 0.03) than LMI (0.39 ± 0.03). Scan statistics showed the lowest agreement (0.18 ± 0.20), specificity (0.93 ± 0.09), sensitivity (0.19 ± 0.27), and high but less reliable accuracy (0.81 ± 0.06) due to over-classifying non-significant areas.

### Conclusion

The analysis underscores how detection methods affect hotspot identification, with LMI as the most reliable method and kriging providing complementary insights. Low sensitivity values indicate the need for a high-quality dataset to ensure more reliable and comprehensive outcomes. Spatial fuzzy Kappa statistics proved effective by integrating the spatial structure of neighbouring wards and random comparisons for more robust statistical significance, highlighting their value for spatial epidemiology. These findings can support public health officials in refining surveillance strategies and targeted interventions.

### Keywords

Spatial clustering ; Spatial fuzzy Kappa index ; Getis-Ord Gi* ; hotspot ; dengue.

# KEEPING BALANCE: GENOMIC SURVEILLANCE OF SHIGA TOXIN-PRODUCING *ESCHERICHIA COLI* AND SUSTAINABLE RESOURCES.

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Genomic surveillance of Shiga toxin-producing *Escherichia coli* (STEC) through whole-genome sequencing (WGS) is the state-of-the-art tool to identify clusters and to fully characterize circulating strains enabling risk assessment for haemolytic uraemic syndrome (HUS) development. Also, WGS facilitates national and international data exchange. From 2019 to 2024, the National Reference Centre (NRC) for STEC performed WGS on all strains (n=1207) including only one isolate per familial cluster. Yet, a sustained increase in the number of STEC strains, non-O157 in particular, was observed after the COVID-19 pandemic (2019: 122; 2020: 78; 2021: 118; 2022: 179; 2023: 330; 2024: 404). Therefore, starting from 2025, WGS performance criteria will be implemented to make genomic surveillance sustainable. The following STEC isolates will be selected for WGS:

1. All STEC-HUS isolates will be sequenced as HUS cases can be considered as the "tip of the iceberg", used as radar for the epidemiologic surveillance of STEC. From 2019 to 2024, 129 out of 1,231 strains (10.5%) were isolated from HUS patients. 4.6% of all STEC-HUS isolates showed *stx1*, 81.4% *stx2* and 14.0% *stx1* and *stx2*. Subtype *stx2f* was not identified among the STEC-HUS isolates.

2. Isolates positive for *stx2* (except *stx2f*) will be sequenced as *stx1* is less prevalent (4.6%) in STEC-HUS isolates and *stx2f* even absent.

3. Isolates belonging to the two major serogroups O26 and O157 will be sequenced as they are important causes of HUS, respectively, 34.1% (n=44) and 26.6% (n=34).

4. Isolates (possibly) involved in an outbreak. Daily monitoring of characteristics, obtained by traditional typing, and geographic location of isolates is done to timely identify suspicious clusters.



Applying these WGS performance criteria to the 2024 numbers (n=404), the number of sequenced strains would be reduced by 34.9%, excluding 95 *stx1*-positive and 46 *stx2f*-positive strains. This will optimize utilization of resources at the level of the NRC but also focus efforts of public health agencies on patients infected with the most pathogenic STEC.

## Keywords

STEC ; Genomic surveillance ; Whole-genome sequencing ; Sustainable resources ; National Reference Centre

# THE EVOLVING EPIDEMIOLOGY OF LOUSE-BORNE *BARTONELLA QUINTANA* IN CANADA.

C. Boodman[1].

1 Institute of Tropical Medicine, Antwerp, Belgium.

## Background

The epidemiology of louse-borne *Bartonella quintana* is poorly known in Canada. This study aims to summarize the epidemiologic trends of *B. quintana* in Canada using vector and human data.

## Methods

This study combines an analysis of laboratory data from Canada's National Microbiology Laboratory (NML) with a systematic review of the literature and a prospective body lice study. Laboratory data included quantitative polymerase chain reaction (qPCR) cycle threshold values and indirect immunofluorescent antibody (IFA) titers. For the systematic review, we searched PubMed, Scopus, Embase, and Web of Science for articles published before 15 July 2024, with terms related to *B. quintana* in Canada.

## Results

Thirty-three individuals with qPCR-positive *B. quintana* were documented in seven provinces and one territory. The number of cases increased over time ($p$-value = 0.005), with the greatest number of cases being reported in 2022 and 2023. The percent positivity for the *B. quintana* qPCR performed at the NML increased over time ($p$-value = 0.036). The systematic review identified fourteen individuals with qPCR-positive *B. quintana* and seven probable cases of *B. quintana* disease. Four of these twenty-one individuals from the systematic review died (19%). All fatalities were attributed to endocarditis. A cluster of transplant donor-derived *B. quintana* infection occurred in 2021-2023. Of 11 *Bartonella* seropositive donors, 6 recipients developed *B. quintana* disease. 556 body lice were analyzed from 7 individuals. 17 louse pools (218 lice) from 1 person were positive for *B. quintana*.

## Conclusion

The detection of *B. quintana* disease in seven provinces and one territory suggests that *B. quintana* has a national distribution in Canada with dozens of urban and rural foci of transmission. *B. quintana* disease is increasingly diagnosed in Canada with novel routes of transmission from solid organ transplant donors and concealed transmission from body lice.

## Keywords

Vector ; bacteria ; homelessness ; lice ; endocarditis.

# BE-MOMO IN NURSING HOMES (BELGIAN MORTALITY MONITORING): METHODOLOGY, INSIGHTS, AND ADDED VALUE.

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## Background

Nursing home (NH) residents are among the most vulnerable groups during public health emergencies, such as the COVID-19 pandemic, seasonal influenza outbreaks, or extreme environmental events like heatwaves. To address these challenges, the Belgian Mortality Monitoring (Be-MOMO) project at Sciensano was expanded in 2024 with a dedicated surveillance system specifically for NH residents, known as 'Be-MOMO in NH'. This system enables near real-time detection and quantification of excess mortality at national and regional levels, stratified by age group (65-84, 85+) and sex.

## Methods

Be-MOMO in NH relies on weekly mortality data provided by Statbel, which identifies NH residents using a proxy derived from household position codes in the National Register. The statistical model Be-MOMO, first developed in 2004 and refined over the years, is applied to this data to identify deviations from expected mortality patterns, by comparing observed deaths to expected mortality based on historical trends of the past five years.

## Results

Our analyses revealed significant findings, such as the absence of excess mortality within NHs during Belgium's third COVID-19 wave, despite its prevalence in the general population. Additionally, the proportion of NH residents deaths among total mortality exhibited a clear seasonal pattern prior to the pandemic, which has since been disrupted.

## Conclusion

This new surveillance system not only enhances situational awareness but also informs public health decision-making by enabling early detection of mortality trends related to public health emergencies. The system achieves these outcomes in a sustainable way by leveraging existing data streams without burdening NHs or regional authorities.



## Keywords

Be-MOMO ; mortality ; surveillance ; nursing homes ; modelling.

# COMPARATIVE GENOMIC ANALYSIS OF MRSA STRAINS ISOLATED FROM HUMAN AND VETERINARY SAMPLES.

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## Background

Methicillin-resistant *Staphylococcus aureus* (MRSA) is a multidrug-resistant organism involved in human and animal infections, but less is known regarding the transmission of pathogenic strains between humans and companion animals. The aim of this study was to compare MRSA isolated from companion animals and humans.

## Methods

MRSA strains were isolated from cats and dogs. These strains were subjected to whole-genome sequencing and compared to previously characterised strains isolated from humans during the national hospital MRSA surveillance of 2022-2023 regarding Multilocus Sequence Typing, resistome and virulome.

## Results

133 human strains and 64 animal strains were analysed. Among them, Among them, 64 strains (51 human, 13 animal) belonged to the clonal complex (CC) CC5, 36 (23 human, 13 animal) belonged to the CC8 and 30 (21 animal, 9 human) belonged to the sequence type (ST) ST398 Methicillin-resistant was conferred by *mecC* in 3 animal strains, while conferred by *mecA* for all other strains. *grlA, grlB* or *gyrA* mutations conferring resistance to quinolones were identified in 74.8% of human strains and 85.9% of animal strains. *tst*, involved in the toxic shock syndrome in humans, was identified in 11 human strains and 11 animal strains. However, while *tst*-harbouring strains were mainly from ST22 in humans (n = 7), they were mainly from ST8 in animals (n = 6).

## Conclusion

This study demonstrated that certain MRSA lineages have the capacity to infect or colonise both companion animals and humans. Notably, CC5 and CC8 commonly found in human infections were also isolated from companion animals, while ST398, previously associated with livestock-associated MRSA infections, was commonly found in humans and in animals.

## Keywords

Methicillin-Resistant *Staphylococcus aureus* ; Humans ; Companion Animals ; Multilocus Sequence Typing ; Whole Genome Sequencing.

# FIRST WINTER SEASON OF SARS-COV-2, INFLUENZA AND RSV MONITORING IN BELGIAN WASTEWATER.

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## Background

In Belgium, a national wastewater-based surveillance program was established in 2020 to monitor SARS-CoV-2. In September 2024, this program was extended to two other respiratory viruses: RSV A/B and influenza A/B virus. Challenges encountered to adapt the existing surveillance program will be discussed in this work.

## Methods

Over 5 million inhabitants representing 40% of the Belgian population are monitored throughout the weekly sampling of 30 wastewater treatment plants. Analysis of RSV A/B, influenza A/B, SARS-CoV-2 viral concentrations are performed by RT-ddPCR. On a weekly basis, epidemiological assessments are conducted using wastewater-based indicators, and weekly assessments are published by the Belgian Public Health Institute, Sciensano.

## Results

A retrospective study performed on samples stored between January and April 2024 demonstrated that lower viral concentration were measured for RSV A/B and influenza A/B than for SARS-CoV-2. Therefore, two distinct ways for analysing data were developed. Firstly, the proportion of data above the limit of quantification being high for SARS-CoV-2, wastewater-based indicators were developed based on quantified measurements. Secondly, the low proportion of RSV and influenza measurements being above the limit of detection lead to the development of another reporting system. The number of treatment plants where a target was detected were aggregated by date.

## Conclusion

Since January 2024, two of the three respiratory viruses studied, SARS-CoV-2 and Influenza A/B, were effectively monitored through wastewater-based surveillance at a national level. And, an entire winter of observation is necessary to conclude on the effectiveness of wastewater-based epidemiology to monitor RSV A/B. These data were collected during winter 2024/2025 and will be presented at the Symposium in May 2025.

## Keywords

Wastewater-based epidemiology ; SARS-CoV-2 ; RSV ; Influenza ; alerting indicator.

# MOLECULAR SURVEILLANCE OF MEASLES TRANSMISSION IN ELIMINATION SETTINGS.

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## Background & Aim

Molecular measles surveillance is paramount to track measles transmission within a population. Due to worldwide efforts to reach measles elimination, only genotypes B3 and D8 are circulating, resulting in a low genetic variability. In 2020, Belgium has been declared "measles eliminated" and consequently, tracking measles transmission based on nucleocapsid-gene (N450) sequencing is challenging. We evaluated the usefulness of sequencing the non-coding region between the matrix and fusion genes (MF-NCR), in tracking measles transmission in Belgium.

## Methods

Measles samples from 2019 and 2023 received at the National Reference Center for Measles, Mumps and Rubella were sequenced for N450 and MF-NCR. Phylogenetic trees based on N450 and MF-NCR were constructed to compare the phylogenetic clusters obtained with both sequencing approaches. Epidemiological links between cases were evaluated with the probabilistic model described by Penedos *et al*.

## Results

The epidemiological clusters were not supported by the molecular data obtained with N450 sequencing, due to the low genetic variability within this region. The genetic variability obtained with MF-NCR sequencing is higher, resulting in a better association between the epidemiological and phylogenetic clusters. Due to practical difficulties, only a limited number of epidemiological links could be evaluated with the model of Penedos *et al*. In 2019, several epidemiological links could be excluded based on MF-NCR sequencing while only one epidemiological link could be excluded in 2023.

## Conclusion

MF-NCR sequencing is a promising tool to enhance the resolution of molecular measles surveillance. Before implementing this approach in tracking measles transmission, practical aspects need to be improved. For now, MF-NRC sequencing can be a useful tool in addition to N450 sequencing to assess endemic circulation in retrospect.

## Keywords

Measles ; Elimination ; Genotyping ; Surveillance ; Sequencing.

Presented at 12th ANNUAL BELVIR Symposium.

# MOZAIC – PIECING TOGETHER COMPLEX DATA INTO A CLEAR PICTURE.

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## Background & Aim

The COVID-19 pandemic has significantly impacted Belgium and other European nations, highlighting the need for a multidimensional approach in crisis management. Effective decision-making during crisis requires integrating diverse datasets to construct a comprehensive situational analysis. To address this challenge, we developed MOZAIC (Multidimensional Overlay for Zonal Assessment and Informed Choices), a tool designed to consolidate and visualize global and local multisectoral data thus supporting data-driven decision-making.

## Methods

MOZAIC is a Shiny application developed using open-source technologies, including R and Leaflet. It enable users to explore predefined map layers or generate custom ones with their own datasets. MOZAIC supports various file formats, such as CSV, XLSX and GeoJSON, ensuring compatibility with widely used data sources.

The key features include:

- Layer Management: Users can select, overlay, and rearrange different layers to analyze complex scenarios across multiple dimensions.
- Data Wrangling: MOZAIC provides options to filter and manipulate variables within datasets.
- Diverse Visualization: Several data representation methods are supported, including choropleths, markers, lines, circles, and various charts.

## Results

MOZAIC enhances crisis response by integrating (but not limited to) epidemiological, social, environmental, and economic data into a single, interactive platform. The application is currently tested, demonstrating its ability to:

- Facilitate rapid identification of high-risk zones and vulnerable populations.
- Support scenario-based decision-making by enabling dynamic data overlay and comparison.

## Conclusion

MOZAIC represents an advance in crisis management by strengthening decision-making processes in complex emergency situations. Future developments will focus on enhancing time-series data visualization by incorporating animated map representations and refining the tool based on user feedback.

## Keywords

Crisis management ; Decision-making ; Shiny Application ; Data Integration ; Geospatial Visualization.

# NEW DEVELOPMENTS IN NANOPORE METAGENOMIC SEQUENCING FOR AMR SURVEILLANCE: A CASE STUDY ON FLUOROQUINOLONE RESISTANCE.

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## Background

Antimicrobial resistance (AMR) is a growing global health threat, necessitating comprehensive surveillance strategies. Traditional methods like phenotypic testing and whole-genome sequencing of cultured isolates remain valuable, but have limitations. Culture-free approaches, such as metagenomic sequencing can offer insights on a broad spectrum of microorganisms and resistance types, including those difficult to isolate. This study aims to demonstrate how Oxford Nanopore long-read sequencing can overcome current limitations of short-read metagenomic sequencing, enhancing AMR monitoring.

## Methods

Using chicken faecal samples, we investigated fluoroquinolone resistance as a case study. We employed Oxford Nanopore long-read sequencing and developed novel strategies to analyze the data. These strategies included linking resistance genes to their bacterial hosts through DNA methylation detection and applying advanced bioinformatic methods for strain-level analysis.

## Results

Our approach successfully linked resistance genes to their bacterial hosts based on DNA methylation detection, and enabled detailed phylogenomic comparisons. We identified fluoroquinolone-resistance point mutations directly from complex metagenomic datasets. These findings demonstrate the potential of long-read sequencing to overcome persistent challenges in AMR surveillance.

## Conclusion

Despite some technical challenges, our study highlights how advances in metagenomic sequencing technology and data analysis can significantly enhance AMR surveillance across diverse environments. This work provides practical insights for integrating these techniques into existing monitoring frameworks to better track, understand, and combat the spread of antimicrobial resistance.



## Keywords

Metagenomics ; Antimicrobial resistance ; Nanopore sequencing ; Strain-resolved metagenomics ; Plasmids.

# DATA QUALITY ASSESSMENT OF THE NEW INFECTION BAROMETER FOR GENERAL PRACTITIONERS IN BELGIUM.

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## Background & Aim
Enhanced infectious disease surveillance in primary care benefits both general practitioners (GPs) and health authorities. The GP Infection Barometer, an extension of the GP COVID-19 barometer, is a new syndromic surveillance system that automates data extraction from Electronic Patient Records (using the ICPC-2 coding system), hence ensuring a low reporting burden. It aims to provide real-time data on 50 infectious diagnoses, including acute respiratory infections, gastrointestinal infections, vaccine-preventable diseases, scabies and impetigo, across 10 age groups. This abstract describes the data quality assessment.

## Methods
Developed in 2023-2024, data collection began on 16/10/2024. The data of the first 61 days were analyzed for quality and consistency through validation checks. These checks assessed uniqueness of record names and practice IDs, timeliness of data submissions, format consistency, missing values, numeric ranges and outliers, stability of practice characteristics over time, accuracy of coded and derived values, the daily rate of coded diagnoses, and participation rate.

## Results
The validation dataset included 47,212 records from 1,219 GP practices using 1 software package. The participation rate exceeded expectations. Validation confirmed overall good data quality and consistency in most checks. Limitations included 1) gaps in timely reporting, requiring further adjustments by the software vendor, 2) low coding rates in some practices limiting data usability and 3) reliance on only 1 of Belgium's 7 GP software vendors, which could hamper long-term sustainability.

## Conclusion
The good quality of the GP Infection Barometer data will enable to enhance primary care surveillance by providing real-time, actionable data for outbreak detection and healthcare planning. However, coding practices and extension to other softwares require further attention.

## Keywords
primary care ; infections ; syndromic surveillance ; extraction ; electronic patient records

# VALIDATION OF A CODE-BASED DATA COLLECTION METHOD IN PRIMARY CARE FOR INFLUENZA-LIKE ILLNESS SURVEILLANCE IN BELGIUM: PROTOCOL.

M. Nahimana[1], S. Bensemmane[1], F. Rouvez[1], L. Debouverie[1], S. Moreels[1] , R. De Schreye[1], N. Bossuyt[1].

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## Background

In Belgium, the COVID-19 Barometer in General Practices (cBGP) rapidly provided COVID-19 data during the pandemic. This code-based semi-automated tool also captured daily data on Influenza-like Illness (ILI). Meanwhile, the long-running questionnaire-based ILI surveillance of the Sentinel General Practitioners network (SGP) is hampered in its expansion.

This study aims to determine gains and losses of replacing the questionnaire-based method with a code-based method for ILI surveillance data collection in Belgian general practices.

## Methods

The Center for Disease Control and Prevention (CDC) guidelines for evaluating surveillance systems will serve as a framework for a retrospective comparison of cBGP with established ILI data collection methods.

First, requirements for the ILI surveillance system in Belgium will be defined. Then, the surveillance systems will be evaluated based on nine attributes: Data quality, ILI cases, Sensitivity, Representativeness, Timeliness, Stability, Simplicity, Acceptability and Flexibility.

Quantitative and qualitative measures, as well as thresholds, will be determined to ensure a comprehensive performance evaluation.

The Simple Multi-Attribute Rating Technique (SMART) method will be applied to identify the optimal approach among three defined alternatives: to retain, complement, or replace the SGP data collection method, outlining focus points. Experts will score and assign importance-related weight to attributes per alternative. The alternative receiving the highest endorsement will be recommended.



## Results

The results will allow the appraisal of the shift from questionnaire- to code-based ILI data collection method in Belgium and highlight key points for optimising cBGP use.

## Conclusion

This assessment will improve the understanding and strengthening of ILI surveillance data collection methods in Belgian primary care.

## Keywords

Influenza ; surveillance ; electronic medical records ; primary care ; validation.

# MOLECULAR EPIDEMIOLOGY OF BOTH INVASIVE AND CARRIAGE *HAEMOPHILUS INFLUENZAE* STRAINS CIRCULATING IN BELGIUM IN 2022.

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## Background

The current study aims to compare the molecular profiles, as determined by whole genome sequencing, of both invasive *Haemophilus influenzae* strains transmitted to the National Reference Center during the early stages of 2022, and carriage strains collected by 40 national laboratories during the same time-period in the frame of a national survey.

## Methods

A total of 127 carriage and 46 invasive strains were sequenced and analyzed for in depth investigation of their resistome, virulome and multilocus sequence typing (ST).

## Results

The clonal complex CC3 was the predominant CC in both invasive (17.4%) and carriage strains (15.8%). Among invasive strains, 13.0% belonged to CC107 and 10.9% to CC11, whereas both CC represented only 4.7% of carriage strains. CC155 and CC422 (6.3% each) were the second most prevalent CCs among carriage strains and were not found among invasive strains. In addition, 16.5% of carriage strains (n = 21) exhibited an unknown ST and were submitted for ST assignment. The proportion of strains exhibiting mutations in the *ftsI* gene, associated with reduced beta-lactam susceptibility, was similar in both groups (invasive 26.1%, carriage 22.1%). The presence of acquired resistance genes associated with ampicillin resistance (*blaTEM-1B*) was detected in about 15% of invasive and carriage strains. An insertion of five amino acids in FolP, which confers resistance to cotrimoxazole, was identified in 15.2% of invasive strains (n=7) and 9.5% of carriage strains (n=12). Furthermore, it was observed that six out of the seven resistant invasive strains belong to CC107 (85.7%). *teltB*, conferring resistance to tetracycline was observed in 4.4% and 1.6% of invasive and carriage cases, respectively.

## Conclusion

CC3 was predominant in both invasive and carriage strains. The proportion of mutations, acquired resistance genes, or insertions associated with antimicrobial resistance were comparable in both groups.

## Keywords

*Haemophilus* ; invasive infection ; WGS ; resistome ; carriage.

# EVALUATION OF THE PATHOGENIC ROLE OF *CAMPYLOBACTER UREOLYTICUS* IN PAEDIATRICS: A MULTICENTRIC RETROSPECTIVE STUDY IN BRUSSELS, 2018-2023.

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## Background

Campylobacteriosis is the leading cause of bacterial gastroenteritis worldwide. Over the past decade, *C. ureolyticus* infections have increased significatively, yet their pathogenicity is debated. This study aims to compare the clinical and biological characteristics of *C. ureolyticus* and *C. jejuni* gastrointestinal infections in the paediatric population.

## Methods

A retrospective review of all paediatric cases of the UHC Saint-Pierre and the University Hospital Of Children Queen Fabiola that had positive cultures for *C. ureolyticus* or *C. jejuni* between 2018 and 2023 was conducted.

## Results

A total of 26 cases of *C. ureolyticus* and 205 cases of *C. jejuni* were reported. Demographic analysis shows few differences between populations, except for a slightly older age distribution in the *C. ureolyticus* group (5.6 ± 0.5 y versus 3.9 ± 0.2 y, $p < 0.005$). Abdominal pain and vomiting were observed at similar rates in both populations, while pyrexia and diarrhea were significantly less common in *C. ureolyticus* cases (fever 17% *versus* 79%; diarrhea 58% *versus* 94%; $p < 0.0005$). Interestingly, chronic diarrhea was more frequently reported in *C. ureolyticus* infections (36% *versus* 10%, $p < 0.005$). *C. ureolyticus* infections were less likely to be associated with elevated CRP levels (31% *versus* 82%; $p < 0.005$) or hematochezia (21% *versus* 77%, $p < 0.0005$). Antibiotic resistance profiles also varied between the two species : *C. ureolyticus* showed notably higher resistance to erythromycin (19% *versus* 1%, $p < 0.005$).

## Conclusion

Although less pathogenic than *C. jejuni*, *C. ureolyticus* can nevertheless induce acute gastroenteritis in pediatric patients. However, it seems less likely to induce inflammatory syndrome, one of the main criteria for therapeutic decisions. Furthermore, genomic analysis will be needed to determine whether *C. ureolyticus* infections are due to several strains with different virulence profiles.



## Keywords

Campylobacteriosis ; *C. ureolyticus* ; Pathogenicity ; Paediatrics ; multicentric study.

# EMERGENCE OF MARIBAVIR RESISTANCE IN TRANSPLANT RECIPIENTS TREATED FOR REFRACTORY/RESISTANT CYTOMEGALOVIRUS INFECTION IN BELGIUM.

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## Background

Maribavir (MBV), Livtencity, is an inhibitor of the cytomegalovirus (CMV) UL97 protein kinase (PK). The drug is indicated for treatment of CMV infection and/or disease refractory to standard antiviral therapies, including (val) ganciclovir (V)GCV, cidofovir, or foscarnet, with or without resistance (R) in adult hematopoietic stem cell transplant (HSCT) or solid organ transplant (SOT) recipients. Maribavir has a favorable safety profile, though there is a concern about its low genetic barrier to development of drug-resistance. A viral rebound under MBV therapy is highly suggestive of emergence of resistance.

## Methods

Under our translational research platform RegaVir (www.regavir.org), we performed CMV genotyping (Sanger sequencing of the UL97 PK, UL54 DNA polymerase, and UL27 genes) to detect emergence of drug-R in 100 samples recovered from 46 transplant recipients receiving MBV for refractory/resistant CMV infection in Belgium.

## Results

Sixty-five samples had a CMV wild-type genotype, and 35 showed mutations associated with MBV-R, including 10 samples bearing changes in the UL97 protein kinase conferring cross-resistance to GCV. MBV-R was detected in 18/46 patients during their follow-up (4 patients with MBV-R/GCV-R infection and 14 with MBV-R). Baseline MBV-R was found in 1/10 patients with MBV-R infection.

Three known UL97 PK drug-resistance mutations were detected: H411Y (9 samples) and T409M (22 samples) associated with MBV-R, and C408F (10 samples) linked to MBV-R/GCV-R. In 6 samples, 2 different UL97 PK substitutions were found: C408F + T409M (2 samples) and T409 + H411Y (4 samples).

## Conclusion

Our results showed a high incidence of emergence of MBV-R in Belgium, with MBV-R CMV infection occurring in 39% of the patients (18/46). Therefore, CMV genotyping is highly recommended in patients with a viral load rebound undergoing MBV therapy.

## Keywords

Cytomegalovirus ; Maribavir ; Drug resistance ; Transplantation ; Genotyping

# IMMUNOGENICITY AND SAFETY OF REDUCED-DOSE *VERSUS* FULL-DOSE MRNA COVID-19 BOOSTER (REDU-VAC): A SINGLE BLIND, RANDOMIZED, NONINFERIORITY TRIAL.

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## Background
Regular booster doses are necessary to strengthen and sustain the immune response following vaccination against COVID-19, ensuring continued protection against severe disease. The inclusion of reduced-dose booster vaccines in the vaccination regimen may offer a potential strategy to extend immune responses while mitigating potential safety concerns associated with higher doses.

## Methods
This phase 4, participant-blinded, randomized, multicenter, non-inferiority study evaluates the immunogenicity and safety profile of a reduced-dose (10µg) *versus* a full-dose (30µg) booster of the mRNA vaccine BNT162b2 in a cohort of 128 participants. A *per*-protocol analysis based on the COVID-19 status at baseline (naïve or previously infected) was conducted. Immunological responses were assessed by measuring anti-RBD IgG titers, and neutralizing antibodies (NT50) against Wuhan and omicron BA.1 at day 0 (D0), day 28 (D28) and 6 months (M6) post booster. Non-inferiority analysis was conducted at D28, calculating the geometric mean ratio (95% CI GMR) of the immune responses of the reduced-dose compared to the full-dose.

## Results
Non-inferiority was not demonstrated, neither for anti-RBD IgG, nor for NT50, with a lower limit of the two-sided 95% CI of GMR >0.67. While the full-dose booster elicited higher anti-RBD IgG and neutralizing antibody titers at D28, this difference decreased over time. By M6, antibody titers remained above D0 in naïve individuals but returned to baseline in those with prior SARS-CoV-2 infection. Notably, the reduced-dose booster demonstrated a more favorable reactogenicity profile, with a lower incidence of local and systemic adverse events. Breakthrough infection (BTI) rates were comparable between both groups, and individuals with BTIs exhibited robust increases in humoral responses.

## Conclusion
These findings suggest that reduced-dose boosting may offer a viable strategy to sustain immune protection while optimizing vaccine safety and distribution.

## Keywords
SARS-CoV-2 ; humoral immune response ; booster vaccination ; non-inferiority analysis ; reactogenicity.

## FIRST REPORTED CASE OF TYPE F BOTULISM IN BELGIUM.

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A 65-year-old patient presented to the emergency department with diarrhea and vomiting after consuming expired food. A diagnosis of gastroenteritis was initially made. The patient returned to the emergency department the following day with persistent digestive symptoms and general weakness. The next morning, the intensive care team was called to the patient's bedside due to coma onset. A diagnosis of carbon dioxide narcosis was rapidly established, and the patient required invasive mechanical ventilation. Initially, no specific etiology was identified. Given a favorable initial course, the patient was extubated a few hours after intubation. However, she later developed a recurrence of carbon dioxide narcosis, seemingly due to axial muscle paralysis, necessitating reintubation. A clinical diagnosis of botulism was suspected by the intensive care team. The patient received equine botulinum antitoxin approximately 65 hours after her intensive care unit admission and showed gradual improvement. Routine PCR assays for *Clostridium botulinum* neurotoxins were negative and the diagnosis of botulism was confirmed only through the mouse bioassay, which detected botulinum toxin type F in the patient's stool. This is the first reported case of botulism caused by the rare type F toxin in Belgium and underscores the continued importance of *in vivo* methods for diagnosis despite the availability of molecular methods.

### Keywords
Botulism ; Intensive care ; Mouse bioassay ; Diagnostics ; Case report.

# SETTING UP A SEVERE ACUTE RESPIRATORY INFECTIONS SURVEILLANCE BASED ON HOSPITAL EMERGENCY DEPARTMENTS IN LUXEMBOURG.

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## Background & Aims

During the COVID-19 pandemic, Luxembourg implemented a resource-intensive hospitalization monitoring, requiring hospitals to report COVID-19 related cases to the health authorities. As the health threat declined, the legal framework changed, and the system was discontinued in April 2023. Since 2024 a syndromic surveillance system (SSS), based on data extractions from existing electronic health records (EHR) of the Hospital Emergency Departments (HED) is implemented in one pilot hospital. Surveillance of severe acute respiratory infections (SARI) is an integral part of SSS. Our aim is to describe the new SARI surveillance system in Luxembourg.

## Methods

Relevant SARI surveillance data were identified from EHR routinely collected at HED. A SARI case was defined as a patient hospitalized for >24h with a HED ICD-10 diagnostic code J09-J22 or U07. Descriptive analysis of SARI cases was carried out for the period 01.01.2023-02.03.2025 by age-group and week. The catchment population of the pilot hospital, used to estimate SARI incidence rates, was determined for each five-year age group by calculating the proportion of hospital-admitted cases in the pilot hospital relative to national hospital admissions within the same age group.

## Results

The syndromic surveillance is operational, with 209,800 HED records received between 1.1.2023 and 2.3.2025. In total, 15,823 (7.6%) records had a J09-J22 HED ICD-10 code, and their weekly number followed the same seasonal trends as positive test results reported by laboratories for RSV, Influenza, and SARS-CoV-2 at national level. Among these there were 3,080 (1.5%) SARI cases, and 438 (0.2%) SARI deaths. The estimated SARI incidence rate was 745 per 100,000 population in 2023 and 795 per 100,000 population in 2024.



## Conclusion

SARI surveillance was successfully established in one hospital, based on data collected routinely without posing a burden to the HED staff. The next steps include automating data extraction, expanding surveillance to additional hospitals, and tailoring the SARI case definition to align with the specific data collected at each hospital.

## Keywords

Syndromic surveillance ; Severe acute respiratory infections (SARI) ; Hospital Emergency Departments (HED) ; ICD-10 codes ; Electronic Health Records (EHR).

# *SALMONELLA* SPP. AS A BE.PREPARED CASE STUDY FOR INTEGRATING MICROBIAL GENOMIC AND HEALTH DATA IN PUBLIC HEALTH SURVEILLANCE.

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## Background

Whole genome sequencing (WGS) has proven its value for public health (PH) surveillance, including pathogen typing, characterization and outbreak investigations. It remains challenging to develop a scalable and sustainable infrastructure that integrates multiple datatypes. Using *Salmonella* as a case study, we present be.Prepared's goals to enhance PH surveillance and response.

## Methods

be.Prepared, supported by EU-funding, integrates infectious disease data via interconnected platforms. Laboratory, clinical and epidemiological data from the National Reference Centre (NRC) for *Salmonella* are transferred near real-time to Healthdata.be. Data are standardized into Data Collection Definitions (DCDs), using *e.g.* international standards (SNOMED-CT, LOINC), to ensure interoperability. Microbial genomic data are processed on a cloud-based bioinformatics platform with features such as automated genomic-based cluster detection. The usability platform combines these data for expert analysis. A pseudonymized datawarehouse is available for epidemiological surveillance.

## Results

New applications were developed in Sciensano LIMS to import the specifications and map the data to the DCDs. The usability platform was set up using BIGSdb and complemented with dynamic visualizations via Microreact. A retrospective collection of more than 3,000 isolates was deposited. Furthermore, automated outbreak alerts were implemented to support rapid, data-driven responses.

## Conclusion

Integration of *Salmonella*, frequently underlying foodborne outbreaks, within the be.Prepared infrastructure, strengthens the Belgian surveillance and is a stepping stone for multi-country outbreak investigations. Its scalable and generic design allows for the integration of additional, relevant data and the inclusion of additional pathogens to build capacity for pandemic preparedness.

## Keywords

Surveillance ; Whole Genome Sequencing ; *Salmonella* spp. ; Public Health ; Interoperability.

## MEASLES RESURGENCE IN BELGIUM IN 2024.

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### Background
In 2020, Belgium received the WHO status "measles eliminated". Whereas only 67 cases were reported in 2023, measles activity was high in 2024. This study describes the epidemiology of the 2024 measles resurgence.

### Methods
It is mandatory to report any suspicion of measles infection to the regional health care inspector. Serology occurs in hospital laboratories, but measles PCR is performed by the National Reference Center for measles, mumps and rubella. Genotyping is performed for cluster identification, surveillance or to differentiate post-vaccinal infection. We combined data from mandatory notifications and NRC testing from 2024.

### Results
A total of 531 cases were reported, with the highest burden in the Brussels-Capital Region (n=305), followed by Flanders (n=129) and Wallonia (n=97). Over half of the cases were unvaccinated (54%) and for 21% vaccination status was unknown. Incidence was highest in children <1y (n=55), which are too young to be vaccinated, followed by children 5-9y (n=190) and 1-4y (n=114). Hospitalization was required for at least 120 persons. Genotyping excluded 11 suspected cases as a vaccine-related rash. The main detected genotype was D8, with only 18 cases identified as the other still circulating genotype, B3. The D8 cases could be subdivided into only 4 different strains, of which the "MVs/Victoria.AUS/39.22" was the most prevalent (60% of genotyped cases). The outbreak had a peak in June and died out in July following targeted vaccination campaigns in schools and day-cares, and the start of the summer holidays.

### Conclusion
The resurgence of measles in Belgium was most pronounced in Brussels, where a diverse and mobile population complicates surveillance and follow-up due to the involvement of multiple authorities. During the discussed outbreak, especially young children were infected. If Belgium wants to maintain the measles elimination status, we should aim to maintain a high MMR vaccination rate among children.

### Keywords
Measles ; Outbreak ; Elimination ; Genotyping ; Vaccination.

# DETECTION OF EXTENDED-SPECTRUM BETA-LACTAMASE (ESBL) AND CARBAPENEMASE-PRODUCING ENTEROBACTERALES (CPE) IN NURSING HOME EFFLUENT: A PILOT STUDY.

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## Background

To explore if the frequency and occurrence of AMR in wastewater accurately reflect the human burden, a pilot study was conducted to assess the microbial dynamic and diversity of ESBL and CPE in the wastewater of nursing home over 24 hours.

## Methods

Between October 2024-February 2025, a 24-hour composite and 8 2-hour interval (6 AM to 10 PM) wastewater samples were collected from each of 7 nursing homes. Samples were collected and transported under refrigeration and immediately analysed. Samples were homogenized and enumerated on CHROMagar™ ESBL, CHROMagar mSuperCARBA™, Tryptone Bile Agar (TBX) and TBX with Cefotaxime (CTX) media. A subset of representative colonies were identified using MALDI-TOF MS.

## Results

In the 24-hour composite samples, 45 strains from 10 different species were identified from ESBL media: *Escherichia coli* (n=11), *Citrobacter freundii* (n=11), *Klebsiella pneumoniae* (n=7), *Enterobacter asburiae* (n=5), *Klebsiella aerogenes* (n=3), *Citrobacter amalonaticus* (n=3), *Enterobacter roggenkampii* (n=2), *Citrobacter braakii* (n=1), *Enterobacter bugandensis* (n=1) and *Enterobacter kobei* (n=1). From CPE media, *C. freundii* (n =5) and *K. pneumoniae* (n=2) were identified. In the 2-hours interval samples, 288 strains from 10 species were identified from ESBL media: *E. coli* (n=90), *C. freundii* (n=74), *C. amalonaticus* (n=40), *K. pneumoniae* (n=24), *Enterobacter hormaechei* (n=20), *K. aerogenes* (n= 17), *C. braakii* (n=11), *E. roggenkampii* (n=9), *Klebsiella oxytoca* (n=2) and *E. kobei* (n=1). From CPE selective media, *C. freundii* (n=14), *E. hormaechei* (n=1) and *E. roggenkampii* (n=1) were identified.

## Conclusion

The species composition differed slightly between the two sampling methods: 8 species were shared while 2 were unique to the 24-hour composite samples and 2 to the 2-hour interval samples suggesting that 24-hour composite sampling provides a representative snapshot of microbial diversity.

## Keywords

Wastewater ; Antimicrobial resistance ; Nursing home ; Enterobacterales ; Microbiology.

## BE-SURVID: STRENGTHENING SURVEILLANCE OF INFECTIOUS DISEASES IN BELGIUM.

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### Background

The COVID-19 pandemic highlighted the value of automated data collection systems that can quickly adapt to emerging pathogens. BE-SURVID (2025–2029) aims to establish a resilient and integrated surveillance infrastructure in Belgium. Building on the COVID-19 experience and aligned with national and EU initiatives, the project focuses on strengthening the surveillance of infectious diseases at all levels of the healthcare system.

### Methods

The project makes use of the be.Prepared architecture, developed within the HERA-BE-WGS initiative, to enable secure, pseudonymized data exchange via the Belgian eHealth service as a Trusted Third Party (approved by the Information Security Committee, deliberation no. 22/268). To improve data quality and operational efficiency, automated system-to-system data transfers from general practitioners, clinical laboratories and hospitals will be implemented. Data will be collected according to international standards (such as LOINC and SNOMED-CT) and align with the Belgian eHealth Action Plan 2022-2024.

### Results

In the short term, BE-SURVID will implement automated data collection across key healthcare sectors while addressing technical and legal challenges. Mid-term objectives include improving the use of data linkage and developing sustainable, integrated surveillance strategies. In the long term, the project will provide a robust system that enhances national preparedness and supports EU-wide surveillance of infectious diseases.

### Conclusion



BE-SURVID will strengthen the detection and response capacity to infectious disease threats in Belgium. By promoting automated, high-quality data collection and integrating different health and social sector data sources, the project will increase public health resilience and improve pandemic preparedness through better early detection, investigation and coordinated response.

### Keywords

Surveillance ; Data Linkage ; Preparedness ; Data Collection ; Infectious Diseases.

# EVALUATION OF A MULTIPLEX REAL-TIME PCR ASSAY FOR THE DIAGNOSIS OF VAGINAL CANDIDIASIS: A COMPARISON WITH CONVENTIONAL LABORATORY CULTURE METHODS.

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## Background
Current gold standard for diagnosis of vulvovaginal candidiasis (VVC) is culture, associated with limited sensitivity and time-consuming steps, consequently delaying antifungal treatment. Non-culture-based DNA detection by PCR, can assist in rapid diagnosis and initiation of species-based therapy. Even though analytical sensitivity of PCR has proven to be superior to culture, the clinical meaning of *Candida* PCR positivity needs to be addressed. Here, PCR was evaluated for semi-quantitative detection of *Candida* spp., as routine alternative for culture in context of VVC.

## Methods
368 vaginal swabs were collected from women presenting with- or in treatment for VVC. Samples were first cultured on CHROMID® agar and isolated strains identified with VITEK® MS MALDI-TOF (BioMérieux), whereafter subjected to a PCR assay (Candidosis Real-TM Quant, Sacace Biotechnologies®) on ELITe InGenius®. *Candida* spp. positivity and sub-species determination were reported.

## Results
Culture and PCR yielded concordant results in 293 of 368 samples (79.6%). Kappa coefficients were 0.69 (*C. albicans*) and 0.73 (C. non-albicans). *Candida* species were detected in 148 patients with both assays (118 *Candida albicans* and 30 *Candida* non-albicans). Compared to culture, PCR had a sensitivity and specificity of, respectively, 100% and 77.7% (*C.albicans*) or 96.8% and 94.7% (*C.* non-*albicans*). The lower specificity for *Candida albicans* detection was attributable to the fact that among the culture-negative patients, 74 (20.1% of total cohort) were positive with PCR. *Ct*-value proved to be a significant predictive marker of the fungal load (CFU/mL) determined in culture (AUC=0.900; $p<0.001$. A diagnostic cut-off of $Ct \leq 23.6$ was established by ROC-analysis, with a sensitivity of 83.2% and a specificity of 84.6%.

## Conclusion
This study established PCR as a valid and fast method for diagnosis of vulvovaginal with a significant increased sensitivity. The use of a diagnostic cut-off *Ct*-value can aid in the distinction between colonization and infection.

## Keywords
PCR ; *Candida* ; Culture ; Sensitivity ; Diagnostic cut-off.

# INCREASE IN CIRCULATION OF NOROVIRUS GII.17 AND THE 2025 WINTER SEASON PEAK.

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## Background

Norovirus is a leading cause of acute gastroenteritis outbreaks worldwide. This RNA virus was first identified during a gastroenteritis outbreak at a school in Norwalk, Ohio, in 1968, and was initially named Norwalk virus before being reclassified as norovirus. Since 2011, Sciensano has hosted the NRC for norovirus, providing over a decade of systematic molecular surveillance data in Belgium.

## Results

Historically, genogroup GII has been the most commonly detected in norovirus outbreaks, with the GII.4 strain being the dominant circulating genotype in Belgium and worldwide for nearly a decade. However, in the winter of 2014/15, a novel GII.17 strain emerged as a major cause of gastroenteritis outbreaks in China and Japan, with sporadic cases reported in Belgium in the years that followed. In 2023, an unexpected and significant increase in GII.17 cases was observed in Belgium, neighboring countries, and the USA, with GII.17 overtaking GII.4 for the first time in 2024. Notably, the winter peak of 2025 in Belgium was almost exclusively attributed to GII.17. Similarly, the USA and UK reported exceptionally high norovirus winter peaks driven by this genotype. The Belgian peak of 2025 could be examined in more detail due to the support of the regional platforms of hospitals.

## Conclusion

The emergence of GII.17 as a dominant strain in the winter of 2025 leaves the population susceptible to reinfection by the still-circulating GII.4, which remains distinct from GII.17 and capable of evading existing population immunity. This potential shift in circulating strains suggests a potential for a second winter peak. This development underscores the importance of continuous surveillance to monitor the evolving norovirus landscape and to anticipate potential public health challenges.

## Keywords

Norovirus ; GII.17 ; surveillance ; emerging genotype ; foodborne pathogen.

# EPIDEMIOLOGY AND MACROLIDE RESISTANCE OF *MYCOPLASMA PNEUMONIAE* INFECTIONS IN BELGIUM AND THE GLOBAL CONTEXT.

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*Mycoplasma pneumoniae* is a small, intracellular pathogen causing respiratory tract infections, particularly in children and young adults. Macrolides are the primary treatment, though resistance is driven by point mutations in the 23S rRNA gene. This study characterizes the epidemiology of *M. pneumoniae* infections in Belgium and situates findings within a global context.

Between 2009 and 2024, 848 *M. pneumoniae*-positive samples were received by the National Reference Center (NRC) for respiratory pathogens. *M. pneumoniae* was detected using an in-house quantitative PCR. Macrolide resistance was assessed in 650 respiratory samples using real-time PCR and melting curve analysis and sanger sequencing (n=7). Whole genome sequencing (Illumina MiSeq) was conducted on a selection of strains (2016, n=2; 2024, n=3). Publicly available *M. pneumoniae* sequences (n=355) were retrieved from NCBI for comparative analysis and multilocus sequence typing (MLST) was employed for strain characterization.

Epidemic waves were observed in Belgium in 2014-2015, 2019-2020 and 2023-2024. The median age of infected patients was 19 years old (range: 0-84) with a 54%/46% male/female ratio. No samples were received in 2021-2022, possibly due to COVID-19 non-pharmaceutical interventions which reduced *M. pneumoniae* transmission. The resurgence of infections in Belgium, observed from September 2023 to December 2024, coincided with a global increase of cases. Globally, 15 sequence types (STs) were identified in public sequences, predominantly ST3 (n=158) and ST14 (n=50) which were also observed in Belgium in 2016. In 2024, ST3, ST7 and one novel ST were identified in Belgium. ST3 and ST14 circulated globally, whereas ST7 was primarily detected in Japan. Macrolide resistance was detected in 25/650 (3.8%) strains. Mutations A2063G (n=5), A2064C (n=1) and A2064G (n=1) in the 23S rRNA gene were identified.

Genomic surveillance can help investigate the recent resurgence of *M pneumoniae* in Belgium. The macrolide resistance rate (3.8%) aligns with other European countries.

## Keywords

*Mycoplasma pneumoniae* ; Whole genome sequencing ; Macrolide resistance ; Respiratory infections ; Surveillance.

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